

**SEMESTER S5**  
**VLSI TECHNOLOGY**

|                                            |                 |                    |                |
|--------------------------------------------|-----------------|--------------------|----------------|
| <b>Course Code</b>                         | <b>PCEVT503</b> | <b>CIE Marks</b>   | 40             |
| <b>Teaching Hours/Week<br/>(L: T:P: R)</b> | 3:0:0:0         | <b>ESE Marks</b>   | 60             |
| <b>Credits</b>                             | 3               | <b>Exam Hours</b>  | 2 Hrs. 30 Min. |
| <b>Prerequisites (if any)</b>              | PCECT302        | <b>Course Type</b> | Theory         |

**Course Objectives:**

1. To give knowledge about IC fabrication techniques and VLSI design methodologies.
2. To impart the skill of analysis and design of MOSFET and CMOS logic circuits.

**SYLLABUS**

| <b>Module No.</b> | <b>Syllabus Description</b>                                                                                                                                                                                                                                                                                                                                                                              | <b>Contact Hours</b> |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>1</b>          | Material Preparation- Purification, Crystal growth (CZ process), wafer preparation.<br>Thermal Oxidation- Dry and Wet oxidation, Diffusion- diffusion techniques.<br>Ion implantation -Technique, annealing.<br>Epitaxy: Vapour phase epitaxy and molecular beam epitaxy Lithography-<br>Photo lithographic sequence, Electron Beam Lithography.<br>Etching and metal deposition, CMOS Twin well process | 9                    |
| <b>2</b>          | Moore's law. ASIC design, Full custom ASICs, Standard cell based ASICs,<br>Gate array based ASICs, SoCs, FPGA devices, ASIC and FPGA<br>Design flows, Top-Down and Bottom-Up design methodologies.<br>Logical and Physical design. Speed power and area considerations in VLSI design                                                                                                                    | 9                    |
| <b>3</b>          | CMOS inverters- DC characteristics, switching characteristics, power dissipation , Layout Design rules, Stick Diagram and layout of CMOS Inverter, two input NAND and NOR gates.                                                                                                                                                                                                                         | 9                    |

|          |                                                                                                                                                                               |          |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|          | MOSFET Logic Design -Pass transistor logic, Complementary pass transistor logic and transmission gate logic , realization of functions                                        |          |
| <b>4</b> | Read Only Memory-4x4 MOS ROM Cell arrays(OR,NOR,NAND) Random Access Memory –SRAM-Six transistor CMOS SRAM cell, DRAM –Three transistor and One transistor Dynamic Memory Cell | <b>9</b> |

**Course Assessment Method**  
(CIE: 40 marks, ESE: 60 marks)

**Continuous Internal Evaluation Marks (CIE):**

| <b>Attendance</b> | <b>Assignment/<br/>Microproject</b> | <b>Internal<br/>Examination-1<br/>(Written)</b> | <b>Internal<br/>Examination- 2<br/>(Written )</b> | <b>Total</b> |
|-------------------|-------------------------------------|-------------------------------------------------|---------------------------------------------------|--------------|
| <b>5</b>          | <b>15</b>                           | <b>10</b>                                       | <b>10</b>                                         | <b>40</b>    |

**End Semester Examination Marks (ESE)**

*In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions*

| <b>Part A</b>                                                                                                                                                     | <b>Part B</b>                                                                                                                                                                                                                                                                    | <b>Total</b> |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| <ul style="list-style-type: none"> <li>2 Questions from each module.</li> <li>Total of 8 Questions, each carrying 3 marks</li> </ul> <p><b>(8x3 =24marks)</b></p> | <ul style="list-style-type: none"> <li>Each question carries 9 marks.</li> <li>Two questions will be given from each module, out of which 1 question should be answered.</li> <li>Each question can have a maximum of 3 sub divisions.</li> </ul> <p><b>(4x9 = 36 marks)</b></p> | <b>60</b>    |

**Course Outcomes (COs)**

At the end of the course students should be able to:

| <b>Course Outcome</b> |                                                                | <b>Bloom's<br/>Knowledge<br/>Level (KL)</b> |
|-----------------------|----------------------------------------------------------------|---------------------------------------------|
| <b>CO1</b>            | Summarize MOSFET fabrication techniques                        | <b>K2</b>                                   |
| <b>CO2</b>            | Outline the various methodologies in ASIC and FPGA design.     | <b>K2</b>                                   |
| <b>CO3</b>            | Understand CMOS technology.                                    | <b>K2</b>                                   |
| <b>CO4</b>            | Design VLSI Logic circuits with various MOSFET logic families. | <b>K2</b>                                   |
| <b>CO5</b>            | Compare different types of memory elements                     | <b>K2</b>                                   |

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

**CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)**

|     | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| CO1 | 2   |     |     |     |     |     |     |     |     |      |      | 3    |
| CO2 | 2   |     |     |     |     |     |     |     |     |      |      | 3    |
| CO3 | 2   | 2   | 2   |     | 2   |     |     |     |     |      |      | 3    |
| CO4 | 2   | 2   | 2   |     | 2   |     |     |     |     |      |      | 3    |
| CO5 | 2   |     |     |     |     |     |     |     |     |      |      | 3    |

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

| Text Books |                                            |                      |                       |                            |
|------------|--------------------------------------------|----------------------|-----------------------|----------------------------|
| Sl. No     | Title of the Book                          | Name of the Author/s | Name of the Publisher | Edition and Year           |
| 1          | Introduction to VLSI Circuits and Systems, | John P Uyemura,      | Wiley India,          | 2006                       |
| 2          | VLSI Technology                            | S.M. SZE             | McGraw-Hill           | Indian Edition, 2/e., 2003 |

| Reference Books |                                                      |                                     |                                                  |                      |
|-----------------|------------------------------------------------------|-------------------------------------|--------------------------------------------------|----------------------|
| Sl. No          | Title of the Book                                    | Name of the Author/s                | Name of the Publisher                            | Edition and Year     |
| 1               | Digital Integrated Circuits- A Design Perspective    | Jan M. Rabaey                       | Prentice Hall                                    | Second Edition, 2005 |
| 2               | Principles of CMOS VLSI Design-A Systems Perspective | Neil H.E. Weste , Kamran Eshraghian | Pearson Publication                              | Second Edition, 2005 |
| 3               | Design of Analog CMOS Integrated Circuits            | Razavi                              | McGraw Hill Education India Education, New Delhi | 1e, 2003.            |
| 4               | CMOS Digital Integrated Circuits- Analysis & Design  | Sung –Mo Kang & Yusuf Leblebici,    | McGraw-Hill,                                     | Third Ed., 2003.     |
| 5               | Fundamentals of Modern VLSI Devices                  | Yuan Taur & Ning,                   | Cambridge University Press,                      | 2008                 |

| Video Links (NPTEL, SWAYAM...) |                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Module No.                     | Link ID                                                                                                                                                                                                                                                                                                                                                   |
| 1                              | <a href="https://youtu.be/Iv4Cj2A3ldw?si=8wkrFXV583vMiJDh">https://youtu.be/Iv4Cj2A3ldw?si=8wkrFXV583vMiJDh</a> ,<br><a href="https://youtu.be/fokSc0xITfM?si=pbWd_UtVbvVacPA">https://youtu.be/fokSc0xITfM?si=pbWd_UtVbvVacPA</a> ,                                                                                                                      |
| 2                              | <a href="https://youtu.be/oZSv68esbgI?si=AoGIe2JX9G6GsdYZ">https://youtu.be/oZSv68esbgI?si=AoGIe2JX9G6GsdYZ</a> ,<br><a href="https://youtu.be/4cPkr1VHu7Q?si=aIAxpL5vUn9s24zX">https://youtu.be/4cPkr1VHu7Q?si=aIAxpL5vUn9s24zX</a> ,<br><a href="https://youtu.be/ht7nEjNydDU?si=_E7m1qMrUvqdcya4">https://youtu.be/ht7nEjNydDU?si=_E7m1qMrUvqdcya4</a> |
| 3                              | <a href="https://youtu.be/UuafwIJAKhY?si=PRrPHzB4mWxwm5h7">https://youtu.be/UuafwIJAKhY?si=PRrPHzB4mWxwm5h7</a>                                                                                                                                                                                                                                           |
| 4                              | <a href="https://youtu.be/Y8FvvzcocT4?si=3anWLYD00aACws9y">https://youtu.be/Y8FvvzcocT4?si=3anWLYD00aACws9y</a>                                                                                                                                                                                                                                           |